

JSX Syntax Complete Notes

JSX stands for JavaScript XML. It allows developers to write HTML-like code inside JavaScript while working with React. JSX makes React code easier to read and understand.

1. Why JSX is Used?

Without JSX, React code becomes difficult to read because we must use `React.createElement()`. JSX provides a simpler and cleaner syntax.

```
// Without JSX
React.createElement("h1", null, "Hello");

// With JSX
<h1>Hello</h1>
```

2. class becomes className

In JSX, `class` is replaced with `className` because `class` is a reserved keyword in JavaScript.

```
// HTML
<h1 class="title">Hello</h1>

// JSX
<h1 className="title">Hello</h1>
```

3. for becomes htmlFor

In JSX, `for` is replaced with `htmlFor` because `for` is reserved in JavaScript.

```
// HTML
<label for="email">Email</label>

// JSX
<label htmlFor="email">Email</label>
```

4. Self Closing Tags

All self-closing tags must properly close in JSX.

```
// Wrong


// Correct


<input />
<br />
<hr />
```

5. JavaScript inside JSX using {}

JavaScript expressions can be written inside curly braces.

```
function App() {  
  let name = "Vishvesh";  
  return (  
    <h1>Hello {name}</h1>  
  );  
}
```

6. Inline CSS in JSX

Inline CSS in JSX uses JavaScript objects. CSS properties use camelCase.

```
<h1 style={{ color: "red" }}>  
  Hello  
</h1>  
  
<h1 style={{  
  backgroundColor: "yellow",  
  fontSize: "40px"  
}}>  
  React  
</h1>
```

7. One Parent Element Rule

JSX must return only one parent element.

```
// Wrong  
return (  
  <h1>Hello</h1>  
  <p>Welcome</p>  
)  
  
// Correct  
return (  
  <div>  
    <h1>Hello</h1>  
    <p>Welcome</p>  
  </div>  
)
```

8. React Fragment

Fragments allow multiple elements without adding extra divs.

```
return (  
  <>  
    <h1>Hello</h1>  
    <p>Welcome</p>  
  </>  
)
```

9. Comments in JSX

```
{/* This is JSX Comment */}
```

10. Event Handling in JSX

Event names use camelCase in JSX.

```
<button onClick={fun}>
  Click Me
</button>
```

11. Complete JSX Example

```
function App() {
  let name = "Vishvesh";
  return (
    <div>
      <h1 className="title">
        Hello {name}
      </h1>
      <input type="text" />
      <button
        onClick={() => alert("Button Clicked")}
      >
        Click Me
      </button>
    </div>
  );
}

export default App;
```

12. HTML vs JSX Comparison

HTML	JSX
class	className
for	htmlFor
onclick	onClick
CSS String	CSS Object
<!-- -->	{/* */}
Open tags allowed	Must self-close

13. Important Interview Definition

JSX is a syntax extension for JavaScript used in React that allows developers to write HTML-like code inside JavaScript.